Cooperation discounts are not ratios of raw stage payoffs

Disproof of Conjecture 00000008240

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Abstract

The first assertion of the conjecture identifies the critical discount for cooperation with the ratio of the one-period defection payoff to the cooperative payoff. A standard repeated prisoner's dilemma has a cooperative subgame-perfect equilibrium at discount $1/2$, whereas that ratio is $3/2$. We verify the equilibrium against every unilateral infinite action stream after every finite history. Lean formalizes the stage game, arbitrary pure history-dependent profiles, their infinite discounted payoffs, subgame perfection, and the resulting inequality for the infimum over all cooperative pure equilibria. The example and its equilibrium argument are classical; no new folk theorem is claimed.

The assertion and its interpretation

Both language versions assert the raw-payoff ratio

$$\delta_c = \frac{\text{one-period defection payoff}}{\text{cooperative payoff}}.$$

We interpret cooperation as perpetual mutual cooperation on the equilibrium path, with subgame-perfect incentives under perfect public monitoring. For a given stage game let $\mathcal{S}$ be the set of $\delta \in (0,1)$ for which such a pure history-dependent equilibrium exists, and define $\delta_c = \inf \mathcal{S}$. We will exhibit a member of $\mathcal{S}$ and prove that the set is bounded below, so this infimum is well defined. Any definition allowing mixed equilibria as well has a weakly smaller critical discount and gives the same contradiction.

The source does not specify a noise model or a particular family of static profiles. We do not assign definitions to those later conjuncts: failure of the first conjunct refutes the conjunction. In particular, we do not interpret "defection payoff" as an already normalized payoff *difference*. The standard difference formula $(T-R)/(T-P)$ for grim-trigger incentives is a different expression.

A standard stage game

Consider two players with actions C, D and payoffs

	C	D
C	$(2,2)$	$(0,3)$
D	$(3,0)$	$(1,1)$

Thus $T = 3 > R = 2 > P = 1 > S = 0$. Each player's infinite payoff is $\sum_{n\geq 0} \delta^n u_n$. Multiplying this by $1-\delta$ would leave all incentives unchanged. Every such series converges for $0 \leq \delta < 1$ because $0 \leq u_n \leq 3$.

This is a familiar textbook example, including the exact stage-payoff table in the course handout [2]. The cooperative grim-trigger equilibrium at $\delta = 1/2$ is also Proposition 1 of Levin's notes [1], after adding one to every stage payoff. That addition preserves every discounted payoff comparison.

Theorem 1. *This game has a subgame-perfect equilibrium with perpetual mutual cooperation at $\delta = 1/2$. Consequently $\delta_c \leq 1/2$, whereas $T/R = 3/2$. The first assertion of Conjecture 00000008240 is false.*

Proof. Both players use grim trigger: cooperate until a defection by either player has occurred, and defect forever after that. At any finite history the public state is either c, with no past defection, or p, with punishment already triggered. At $\delta = 1/2$, following the strategy gives continuation values

$$V(c) = 2\sum_{n\geq 0} 2^{-n} = 4, \qquad V(p) = \sum_{n\geq 0} 2^{-n} = 2.$$

For an arbitrary deviator, the stage reward r, next state, and one-step value satisfy the following exhaustive table:

state	own action	r	next state	$r + \frac{1}{2}V(\text{next state})$
c	C	2	c	4
c	D	3	p	4
p	C	0	p	1
p	D	1	p	2

In each case the last entry is at most the current state's value. For any infinite unilateral action stream, induction and telescoping therefore give, for every N,

$$\sum_{n=0}^{N-1} 2^{-n} r_n + 2^{-N} V(s_N) \leq V(s_0).$$

The terminal value is nonnegative. Letting N tend to infinity shows that the deviator's total payoff is at most $V(s_0)$, exactly the payoff from following grim trigger. This argument applies to both players and every finite history, including histories off the cooperative path. It proves subgame perfection against arbitrary pure deviations, not just a single defection. The same pointwise bound also rules out gains from randomized deviations by taking expectations.

Starting from the empty history, both players cooperate forever. Thus $1/2 \in \mathcal{S}$ and $\mathcal{S} \subset (0, 1)$, whence $\delta_c \leq 1/2$. Since the raw defection-to-cooperation payoff ratio is $3/2 > 1/2$, the asserted equality fails. □

Why raw payoff ratios are structurally unsuitable

Adding the same constant to every outcome payoff of a player adds a strategy-independent constant $k/(1-\delta)$ to the repeated payoff. It cannot change any equilibrium or the cooperation threshold. In contrast, the raw ratio T/R changes to $(T+k)/(R+k)$. This is the basic defect in the proposed ratio. The counterexample uses an ordinary strictly ordered prisoner's dilemma and a standard cooperative equilibrium, with no degenerate discount or payoff.

Formal verification and scope

The Lean namespace is `Conjecture8240`. Booleans encode the two actions, and the function `stage` is the first player's actual table; the second player's payoff is the same table with arguments interchanged. Finite histories store both players' actions. A `Profile` assigns an action to each player at every history. The recursive trajectory against a fixed opponent generates the actual history for each unilateral infinite action stream.

The definition `SPE` quantifies over both players, every finite history, a compliant continuation, and every possible unilateral infinite stream. The definition `Cooperates` requires the all-C stream to be compliant from the empty history for each player. `sustainable` ranges over all

such pure profiles at $\delta \in (0,1)$; its infimum is `critical`. Thus the formal critical value is not restricted to grim-trigger profiles.

The central formally verified results are

`history_payoff_summable`, `grim_is_spe`, `grim_cooperates`,
`half_sustainable`, `critical_le_half`, `conjecture8240_false`.

The final theorem negates `ClaimedRatio`, the specialization of the first conjectured equality to this explicit game. All objects needed for the pure-strategy counterexample, including convergence of discounted sums, are formalized. The paper's comments about randomized deviations, the larger mixed-strategy infimum, and invariance under additive payoff shifts are not separately formalized. No exact optimal value for δ_c is needed or claimed here. Only the usual Lean axioms of propositional extensionality, classical choice, and quotient soundness occur in the printed axiom audit.

Source and reproduction

The unmodified bilingual conjecture is included as `SOURCE.md`, retrieved from upstream commit

`b355e42906e6eaf52bc7e9d005e0ef02d8aa7400`.

Its SHA-256 digest and source path appear in `verification/SOURCE_PROVENANCE.md`. The project pins Lean 4.19.0 and Mathlib commit

`c44e0c8ee63ca166450922a373c7409c5d26b00b`.

From the `lean` directory, run `lake build`, or

`lake env lean -DwarningAsError=true Main.lean`.

The work was developed by a delegated AI agent with a self-review; the coordinating agent reviews it separately. There is no external independent review.

References

[1] Jonathan Levin, *Repeated Games I: Perfect Monitoring*, May 2006, Proposition 1. `https://web.stanford.edu/~jdlevin/Econ%20286/Repeated%20Games%20I.pdf`

[2] *Chapter 14: Repeated Prisoner's Dilemma*, Northwestern University course handout discussing Osborne's text, stage game and grim-trigger strategies, pp. 1–2. `https://sites.math.northwestern.edu/~clark/364/handouts/osborne14.pdf`